Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

Lecture 3B – Engineering Anthropometry II

EIN 3241: Human Factors and Ergonomics
Spring 2026

January 30, 2026

Dr. Jad Atweh

Admin Issues

- Quiz 2
  - Wednesday, February 4

Recap

- What is anthropometry
- Design applications
- Measurement devices and techniques
  - Anthropometer
  - Spread Caliper
  - …
- Factors that affect anthropometric data
- Structural dimensions…
  - Stature
  - Acromial Height
  - Eye Height
  - Chest Depth
  - ….

POP QUIZ! WHAT IS THIS CALLED?

Human Variability

- It is variability within and between groups that forces designers to consider adjustability.

- Is there an overall average human?

Correlations of Body Measurements

- Some body dimensions are significantly correlated with each other, while some others are not.
- The higher the correlation coefficient between two body dimensions, the more closely related they are.
- Such data can be used to estimate one dimension from another.

TABLE 5.21 SELECTED CORRELATION COEFFICIENTS FOR ANTHROPOMETRIC DATA, U.S. ADULTS*

	1	2	3	4	5	6	7	8	9	10
1. Age		.223	.048	−.023	.039	−.055	.091	−.072	.233	.287
2. Weight	.113		.533	.457	.497	.431	.481	.370	.835	.799
3. Stature	−.028	.515		.927	.914	.849	.801	.728	.334	.257
4. Chest height	−.028	.483	.949		.897	.862	.673	.731	.271	.183
5. Waist height	−.033	.422	.923	.930		.909	.607	.762	.308	.238
6. Crotch height	−.093	.359	.856	.866	.905		.467	.788	.264	.190
7. Sitting height	−.054	.457	.786	.681	.580	.453		.398	.312	.239
8. Popliteal height	−.102	.299	.841	.843	.883	.880	.485		.230	.172
9. Shoulder circumference	.091	.831	.318	.300	.261	.212	.291	.182		.810
10. Chest/bust circumference	.259	.832	.240	.245	.203	.147	.171	.114	.822	

Example

- If a guy's stature is 177 cm, estimate his:
  - Wrist height
  - Arm reach
  - Hip height

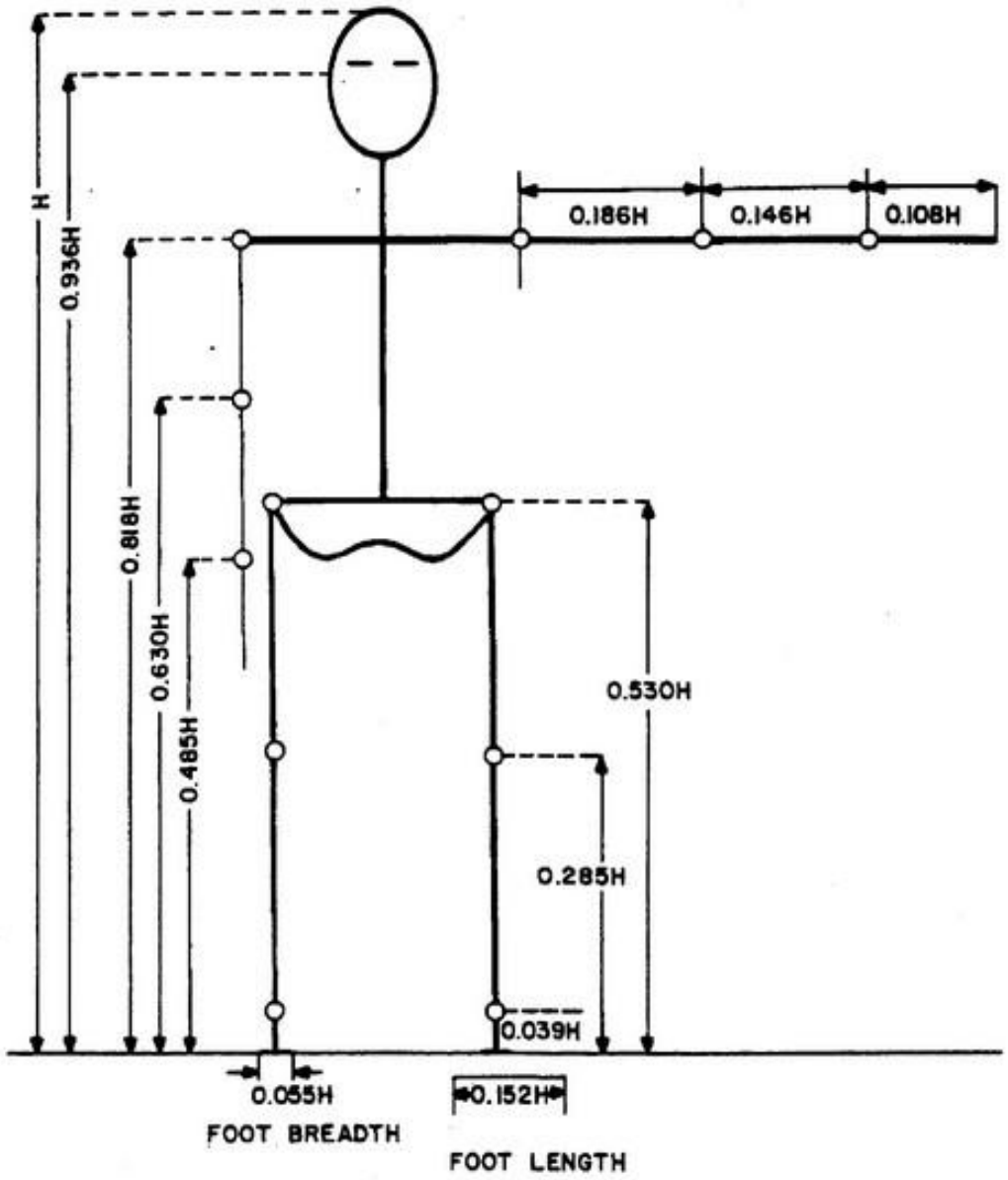
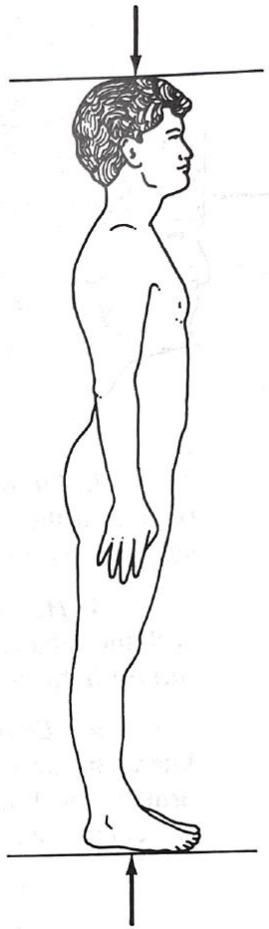


Figure 3.9 Some body segment lengths expressed as proportion of body stature by Drillis and Contini (Roebuck, Kroemer, and Thomson, 1975).

Anthropometry Percentile

Percentiles correspond to the values of a variable below which a specific percentage of the group fall.

POP QUIZ! WHAT IS THIS CALLED?



5th percentile stature for this class is 162 cm, meaning that only 5% of class population is smaller than 162 cm.

50th percentile stature for this class is 173 cm, meaning that 50% of class population is smaller than 173 cm.

95th percentile stature for this class is 185 cm, meaning that 95% of class population is smaller than 185 cm.

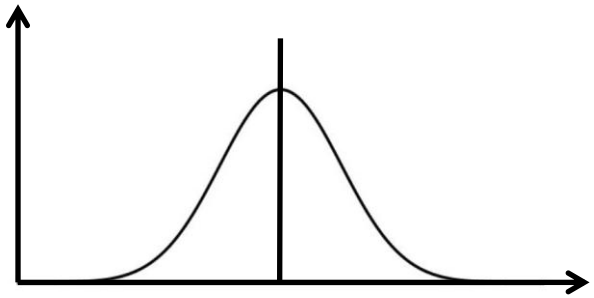
Calculating Percentiles

- **Normal Distribution**: The mean and the standard deviation are two key parameters of normal distribution. The mean (m) is a measure of central tendency that tells us about the concentration of a group of scores on a scale of measurement.

$$m = \frac{\sum x_i}{n}$$

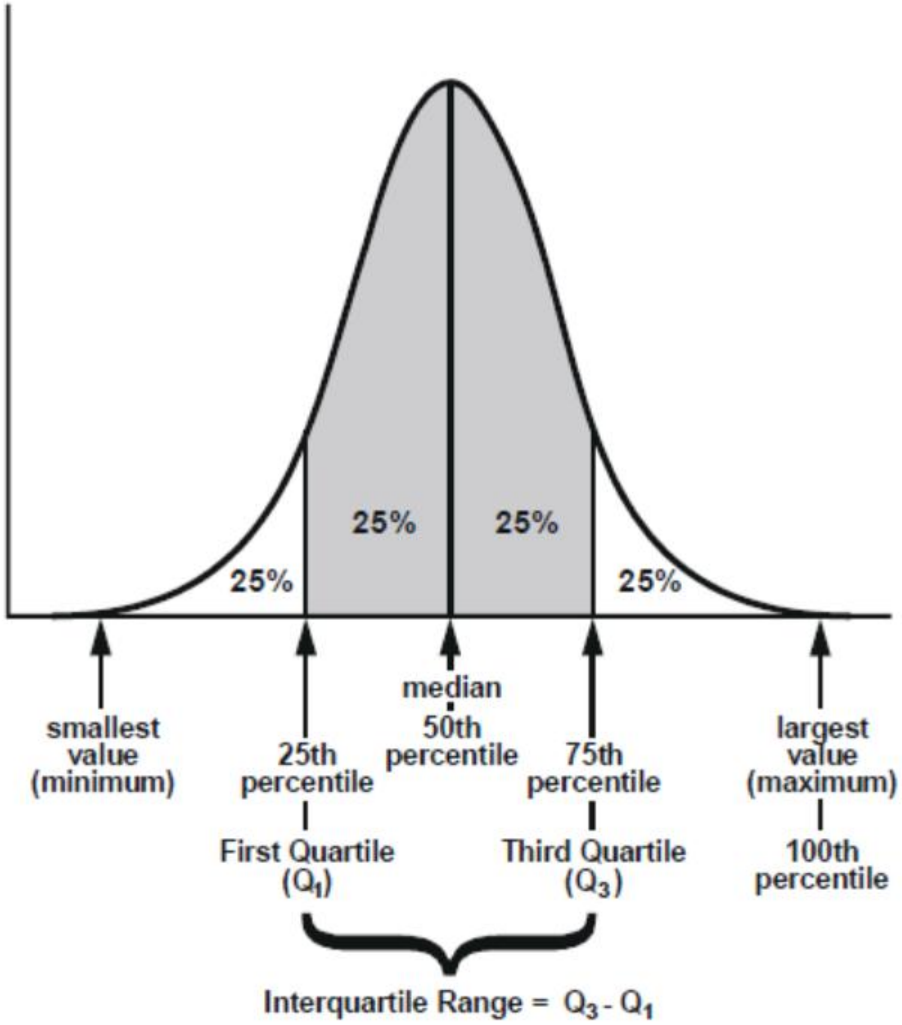
- **The Standard Deviation (SD)** is a measure of the degree of dispersion or scatter in a group of measured scores.

$$SD = \left[\frac{\sum (x_i - m)^2}{(n-1)}\right]^{\frac{1}{2}}$$



Calculating Percentiles

Normal distribution with the quartile distribution



Calculating Percentiles

- Given a set of raw data, the percentile(s) may be found by the following procedure:
  - Find the mean
  - Find the standard deviation (SD).
  - Find the factor F1 (Z) corresponding to a specific percentile point. Multiply this factor by SD and **add to** (if above 50th percentile) or **subtract from** (if below 50th percentile) the mean.

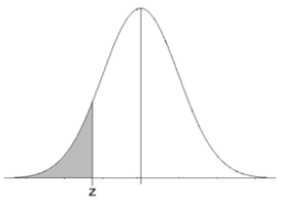
TABLE 5.24
PERCENTILE CALCULATION FACTORS

Percentile	F_1
5th	1.645
10th	1.282
25th	0.674
50th	0
75th	0.674
90th	1.282
95th	1.645

Calculating Percentiles

Normal Distribution Z-table

Standard Normal Cumulative Probability Table



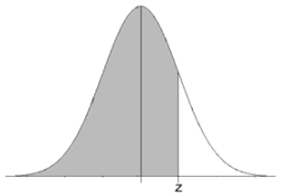
Cumulative probabilities for NEGATIVE z-values are shown in the following table:

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

Calculating Percentiles

Normal Distribution Z-table

Standard Normal Cumulative Probability Table



Cumulative probabilities for POSITIVE z-values are shown in the following table:

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

Another Example

- Calculate the 25th & 75th percentile of the Popliteal Height using the data shown in the following table:

POP QUIZ! WHAT DOES THIS MEAN?

Subject no	Popliteal Height
1	41
2	45
3	40
4	45.5
5	42
6	47
7	44
8	44.5
9	43
10	46

Mean	43.8
SD	2.3

Percentile	z
1St	-2.326
5th	-1.645
10th	-1.282
25th	-0.674
50th	0
75th	0.674
90th	1.282
95th	1.645
99th	2.326

More examples…

	Strategy	Dimensions	Population	Percentile	Special considerations?
Classroom door handle (height)	Average	Elbow height, knuckle height	Student body	50th percentile	Handicap access
Submarine door (height and width)	Extreme	Shoulder span, stature,	Navy personnel	95th percentile	Design of submarine, frequency of use, carrying equipment
Top shelf kitchen cabinet (height)	Extreme	Overhead reach, maybe eye height	General population	5th percentile females	Counter depth, storage needs
Life jacket circumference (ferry)	Range	Shoulder span, waist, neck, chest, & top of arm circumference	General adult population, children, infants	1st to 99th percentile	Extended sizes, children's sizes, infant sizes
Checkout at grocery store (height)	Average	Waist height, elbow height, forward arm reach	General population	50th percentile	Some adjustability for wheelchair users?
Fire extinguisher (trigger force)	Extreme	Hand force/grip force, grip circumference	General population	1st percentile female	May exclude children, older adults

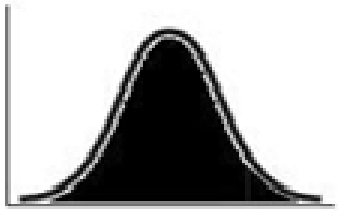
Anthropometric Design Principles

1. Design for the Mean

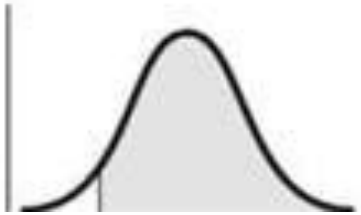
- Design for the mean (50th percentile) user.
- Useful when adjustability is not critical and the task isn't safety-sensitive.
- Risk: Doesn't accommodate users at the extremes.

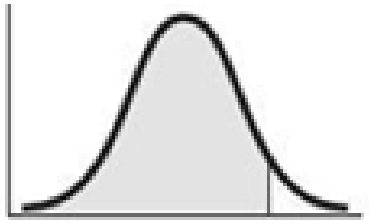
Anthropometric Design Principles

2. **Design for one extreme of spectrum (short or small user)**
   - Workplace design – reach is primary concern



3. **Design for other extreme of spectrum (tall or large user)**
   - Workplace design – ingress and egress is primary concern

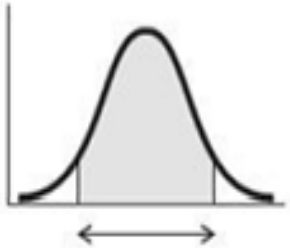




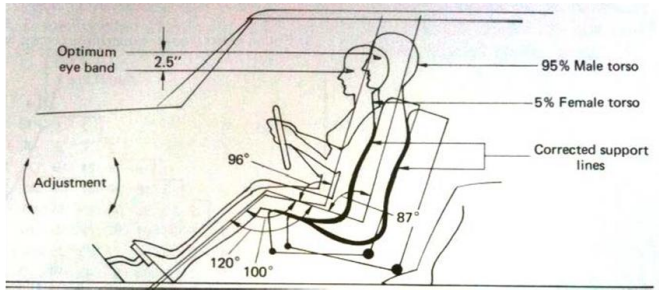
Anthropometric Design Principles

4. Design for adjustability

Provision for using mechanical solution to accommodate a range of users

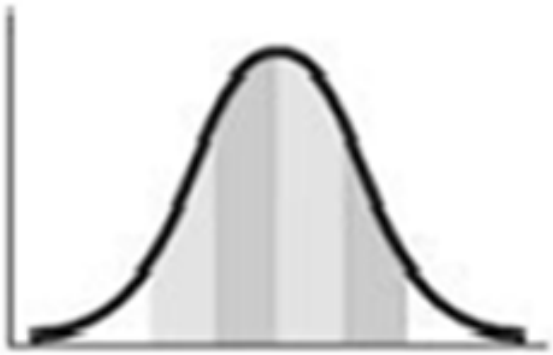


- (the range from the 5th percentile female to the 95th percentile male)
- Design for adjustability:
  - Consumer product – very common
  - Workplaces – not that common
- Not practical or desirable in many design situations:
  - Economically infeasible
  - Design constraints



Anthropometric Design Principles

5. **Design for more types/sizes** for consumer products to accommodate a wide range of population groups



Using Anthropometry in Design

- Determine the applicable body parts
- Determine the user population
  - M, F, children, disabled, pregnant women, etc.
- Determine the percentile to accommodate
- Calculate the appropriate dimension(s)Determine the range of adjustability

Anthropometry Analysis General Approach

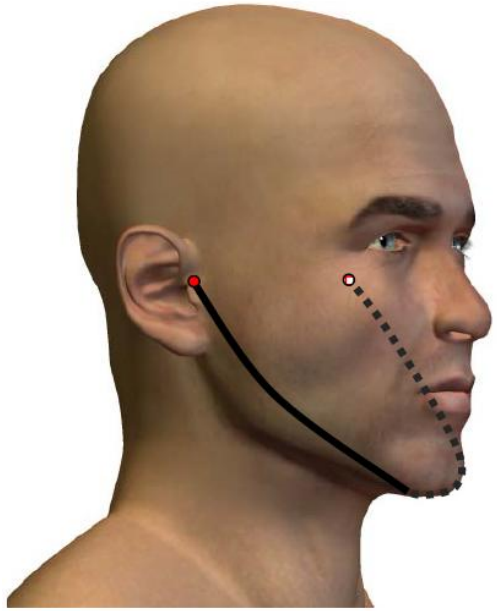
- Step 1: Determine the user population (e.g., children, women, U.S. civilians, age groups, world populations, and races)
- Step 2: Determine relevant body dimensions (e.g., the design of a chair seat must consider waist breadth)
- Step 3: Select the design principles (what percentage of the population is to be accommodated)
- Step 4: Locate the anthropometric tables of the appropriate population.
- Step 5: Determine the percentage of population to be accommodated.
- Step 6: Make appropriate allowances for clothing (not an essential step)
- Step 7: Compute the specific dimension.

What did we learn today?

- Human Variability..
- Anthropometry Percentiles
- Anthropometric Design Principles

Next time…

- Functional Anthropometry
- Examples of Advanced Anthropometry Measures (e.g., ANSUR)
- 3D Anthropometry



UF

Herbert Wertheim
College of Engineering
UNIVERSITY *of* FLORIDA

LEADING THE CHARGE, CHARGING AHEAD